

Effects of an herbal diet, diet x boar line and diet x genotype interactions on skin lesions and on growth performance in post-weaning pigs using a cross-classified experiment

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HIGHLIGHTS

- The highlight of the present study is the analysis of genotype x diet interactions in the context of skin lesions in pigs. This is a novel study, indicating the importance of collaborative work including both disciplines animal breeding and animal nutrition.

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ABSTRACT

The aims of the present study were to assess the effect of an herbal diet (HD; basal diet plus a supplement with chicory herbs) on production traits and lesions scores in post weaning pigs, and to study possible boar x diet and genotype x diet interactions considering the two feeding groups HD and control (CON). In this regard, a cross-classified research design was implemented, aiming on an equal number of boar offspring in both groups HD and CON. Grouping of piglets (993 crossbred piglets from 14 different Piétrain boars sire to German Landrace or German Edelschwein sows) was done after weaning, and the feeding experiment was conducted in post-weaning period up to the pig age of nine weeks. Studied traits included the post-weaning weight (PWW) at the age of nine weeks, average daily gain (ADG) considering the period from weaning until week 9 and a skin lesion score (LS) at week 9. Herbal diet effects and sire x diet effects were inferred via linear models for Gaussian distributed PWW and ADG, and via generalized linear mixed models with a logit link function for categorical LS. Possible genotype x diet interactions were proved in multiple-trait models via Gibbs sampling, by defining same traits recorded either in CON and HD as different traits, i.e., post-weaning weight control (PWW_{CON}) and post-weaning weight herbal diet (PWW_{HD}), average daily gain control (ADG_{CON}) and average daily gain herbal diet (ADG_{HD}), and lesion score control (LS_{CON}) and lesion score herbal diet (LS_{HD}). The dietary treatment had no significant effect ($P > 0.05$) on PWW and ADG. The dietary treatment significantly ($P < 0.05$) influenced the behavior of the pigs, with a lower and favorable LS score for the pigs allocated to HD. Significant sire x diet interactions were identified for LS, PWW and ADG. Pearson correlations between estimates for sire effects from both diets CON and HD (i.e., the sire x diet effect) were 0.84 for PWW, 0.71 for ADG and 0.56 for LS. Genetic correlations smaller than 0.80 between PWW_{CON} and PWW_{HD}, ADG_{CON} and ADG_{HD}, and LS_{CON} and LS_{HD} indicate genotype x diet interactions, especially for the functional and low heritability LS. Hence, re-rankings of sires according to their estimated breeding values (EBV) were observed in both environments CON and LD, displaying smaller genetic and EBV variations for LS in the HD group. The HD dietary supplement contributed to fewer skin lesions, probably due to the favorable effect on pig behavior, but a more pronounced genetic differentiation seems to be possible in the more challenging CON environment. Generally, the results from the present study indicate the importance of genotype x feeding interactions for pig breeding, suggesting specific boars for different feeding environments.

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1. Introduction

In the EU, the 'routinely' use of tail docking is not allowed according to the EU Council Directive 2008/120/EC from December 2008, which focuses on restrictions of non-curative invasions. However, the directive allows tail docking if 'there is evidence that injuries... to other pigs' ears or tails have occurred'. Consequently, about 95% of the pigs in European countries, such as Germany, Denmark, France and others, have docked tails (D'Eath et al., 2014; EFSA, 2007). Nevertheless, the future is to keep pigs with naturally long tails, implying improvements in animal behavior. Aggressive behavior in pigs was observed at different growth stages, especially during the weaning phase (Lau et al., 2015; Schrey et al., 2019). At weaning, major stressors associated with aggressive behaviors include the removal of piglets from the sow, their transfer into a new environment and mixing with unfamiliar pigs, as well as the change from a liquid to a solid diet (Lecce et al., 1979; Campbell et al., 2013). Until clarification of social ranking in the new environment, aggressiveness among pigs causes various behavioral reactions, escalating to damaging fights in addition to tail biting with associated skin lesions (Wurtz et al., 2017; Camerlink et al., 2016, 2014; Turner et al., 2009). Several approaches have been initiated to mitigate aggressive behavior and tail biting through genetic selection, and through modifications of the production environment (Turner, 2011; Rohrer et al., 2013; Ellen et al., 2014). Aikins-Wilson et al. (2021) evaluated prospective breeding strategies in pigs, and identified genetic associations between tail length with tail biting and incidences for skin lesions.

Nutrition played a major role in tail and ear biting, i.e., stimulated foraging, exploratory and social behavior in pigs (Holling et al., 2017; Jensen and Pedersen, 2010; Beattie et al., 2005).

In this regard, feeding diets with low dietary protein levels contributed to an increase of ear and tail biting (e.g. Jensen et al. 1993; Fraser et al. 1991). Some other studies pointed out the favorable effects of fiber levels in the feeding ration on pig behavior (Kallabis and Kaufmann, 2012; Braund et al., 1998; Brouns et al., 1994). Cheng et al. (2017) and Yan et al. (2012) concluded that the inclusion of herbs and their extracts in swine diets improves growth performance, nutrient digestibility, immune function and meat quality. In this regard, the herbaceous plant chicory (*Cichorium intybus* L) has been used as a fiber component in pigs and small ruminant diets, without compromising performance traits and improving nutrient digestibility, immune function and meat quality (Liu et al., 2018; Cheng et al., 2017; Nwafor et al., 2017; Yan et al., 2012; Ivarsson et al., 2011). In addition, active ingredients such as proteins, metabolic energy, vitamins, minerals and bioactive compounds in plant-based diets improved digestion, metabolism and increased overall vitality (Saeed et al., 2017). The nutrition company Dr. Schaette, Saluvet GmbH (personal communication, 2018) indicated the favorable feeding effects of the herbal diet "Kräuterkraft Aufzucht & Verdauung" which consists of 30% chicory root to reduce aggressiveness and simultaneously improving growth performance. However, such hypotheses indicating favorable feeding effects of the herbal diet on pig behavior and performance traits are mostly based on visual inspections and lacking scientific evidence.

Furthermore, the genetic component as well as genotype x diet interactions affected primary and functional trait responses in pigs (Schiavon et al., 2019). Genotype by environment interactions with associated re-rankings of animals in different production environments were mainly due to differences in nutritional characteristics (Hamilton et al., 2003). Augspurger et al. (2002) carried out a feeding experiment in pigs and concluded that different genotypes have different nutrient requirements for growth performances, and differed with regard to feed intake and feed efficiency. Accordingly, Fabian et al. (2002) evaluated growth parameters and reported significant interaction effects for amino acid contents in the diets in different genetic Duroc lines. A selection focus on boar lines was suggested to improve production efficiency and profitability in pig farming systems (Elbert et al., 2020; Vázquez-Gómez et al., 2020; De Cuyper et al., 2019). Especially in rotational

crossbreeding systems (i.e., in crisscross designs) with intra-herd replacements of sows, the genetics of dam lines is less variable than the genetics of end products due to the utilization of bars from worldwide operating breeding companies. Elbert et al. (2020), Vermeulen et al. (2016) and Bidanel and Ducos (1996) studied the growth trajectory in pigs and found strong effects of the boar on birth weight up to weights gains in the fattening period, without evaluating possible genotype by nutrition interactions.

Consequently, the objective of the present study was to implement a cross-classified research design to infer the effects of the herbal diet and of diet x genotype interactions on skin lesions and growth traits in post-weaning pigs. In this context, also quantitative-genetic models were defined for the estimation of variance components and variance ratios in different feeding systems, and for the proof of possible diet x genotype interactions through genetic correlation estimates.

2. Materials and methods

2.1. Animals, housing and feeding

The present study considered 993 crossbred pigs from 77 German Landrace x German Edelschwein dams sired by 14 different Piétrain boars, kept at the teaching and research station "Oberer Hardthof" of the Justus Liebig University in Giessen, Germany. From birth onwards, piglets suckled their mother and additionally received a supplementary diet until weaning at 24 ± 2 days of age. At weaning, piglets were allocated to either a basal diet (CON) or herbal diet (HD) group. The feed ration in the CON dietary treatment included wheat, soybeans and barley as concentrates plus minerals and vitamins. The HD treatment was the basal diet plus a mixture of 0.2% "Kräuterkraft Aufzucht & Verdauung". The composition of the supplementary feed "Kräuterkraft Aufzucht & Verdauung" in the HD diet consisted of 30% chicory (*Cichorium intybus*) root, carbonic acid algae lime (*Lithothamnion calcareum*), alumroot (*Heuchera americana*), 5% yarrow herb (*Achillea millefolium*), fenugreek seeds (*Trigonella foenum-graecum*), nettle herb (*Urtica dioica*), malt sprouts, sugar beet molasses, yeast and seaweed (*Ascophyllum nodosum*) flour. Piglets in both groups CON and HD were fed a two-phase diet after weaning: Phase 1 (starter diet) was fed for 6 days, followed by phase 2 (growing diet) until they are sold for fattening. Piglets were fed ad libitum with a sensor feeding system that supplied diets at appropriate time intervals. Samples of all feed mixtures were collected and analyzed in an accredited laboratory (Landesbetrieb Hessisches Landeslabor, Kassel). Ingredient composition and analyzed

Table 1
Ingredient composition of the experimental diets¹.

Stage Treatment	Starter diet		Growing diet	
	CON	HD	CON	HD
Dry matter (%)	89.8	89.9	88.6	89.4
Metabolic energy (MJ/kg)	14.4	14.4	14.1	13.0
Ingredient, % of DMI				
Insoluble ash	5.2	5.6	8.0	7.9
Crude Protein	18.2	18.8	16.6	17.1
Crude fiber	3.1	3.3	2.4	4.9
Crude fat	6.0	6.3	3.3	3.3
Calcium	0.75	0.60	1.06	1.57
Copper	0.01	0.01	0.01	0.02
Total phosphorus	0.56	0.55	0.52	0.62
Magnesium	0.15	0.14	0.15	0.16
Sodium	0.18	0.16	0.21	0.33
Zinc	0.02	0.01	0.02	0.01
Lysine	1.25	1.31	0.98	1.22
Methionine	0.41	0.39	0.35	0.40
Cystin	0.26	0.28	0.30	0.28
Threonine	0.80	0.79	0.67	0.71

¹CON: basal diet with no additives; HD: basal diet with a mixture of 0.2% "Kräuterkraft Aufzucht & Verdauung".

nutrient content of the experimental diets are presented in Table 1.

Piglets were allocated to the dietary treatment considering a similar number of offspring per boar in each treatment to ensure the same genetic background in both groups CON and HD. Accordingly, 534 and 459 piglets were considered for CON and HD, respectively. Maximal 15 piglets were allocated to 8 pens per treatment, implying a floor space of at least 0.36 m² per piglet. Temperatures were regulated electronically, to ensure a constant climatic production environment from 29 °C at the beginning of the rearing period to 21 °C at the end of the experiment. Each pen was equipped with two automated nipple drinkers at different heights and a one-meter trough.

2.2. Animal traits

Individual piglet weights were determined at weaning at an age of four weeks (weaning weight; WW) and post-weaning age at the age of nine weeks (postweaning weight; PWW). Average daily gain (ADG) in g per day (g/day) was calculated for the period from weaning to the age at nine weeks considering the measurements for WW and PWW. The same pigs were scored for skin lesions at the weaning and at the post-weaning weight date from one trained classifier. In this regard, lesion scores (LS) were assigned based on the protocol described by Pluske and Williams (1996) and Bünger et al. (2015). Accordingly, a lesion implied a single or continuous scratch, and scores were assigned as follows: score 1 = no lesion or less than 5 slight skin lesions; score 2 = mild lesion with more than 5 mild lesions including hair loss, redness, irritation, scratches or small abrasions; score 3 = severe lesion, bleeding and loss of tissue. The descriptive statistics for the traits PWW, ADG and LS is given in Table 2. The pedigree dataset for all animals with phenotypic records for PWW, ADG and LS was traced back to five generations.

2.3. Statistical analyses

2.3.1. Models inferring fixed effects, random effects and sire $\times$ diet effects

Statistical analyses were performed to infer fixed and random effects on performance traits and skin lesions using the statistical software package R (R Core Team, 2020). The “lme4” package (Bates et al., 2015) was applied to fit the linear mixed model (model 1) for the Gaussian distributed traits ADG and PWW as follows:

$$y_{ijklmnopqr} = \mu + L_i + G_j + T_k + LN_l + SL_m + YS_n + b_1(WW_o) + b_2(age_p) + S_q + (S_q \times T_k) + e_{ijklmnopqr} \quad (1)$$

where $y_{ijklmnopqr}$ was the observation for PWW or ADG; μ was the overall mean effect; L_i was the fixed effect of the i^{th} litter size group (five groups: 3–11, 12–15, 16–17, 18–19 or 20–24); G_j was the fixed effect of the j^{th} sex (male or female); T_k was the fixed effect of the k^{th} treatment (CON or HD); LN_l was the fixed effect of l^{th} litter number ($l = 1$ to 5); YS_n was the fixed effect for the n^{th} year-season at recording; SL_m was the fixed effect

Table 2

Descriptive statistics for post weaning weight (PWW), average daily gain (ADG) and lesion score (LS) recorded at the post-weaning date.

Trait	Group ¹	Mean	SD	CV (%)	Min	Median	Max
PWW (kg)	CON	18.94	4.60	0.24	7.70	18.90	31.80
	HD	20.00	4.61	0.23	8.60	19.40	35.00
ADG (g)	CON	358.00	94.18	0.26	90.00	360.00	630.00
	HD	365.86	92.91	0.25	40.00	360.00	660.00
LS					Score	Score 1	Score 2
					0 (in%)	(in%)	(in%)
	CON	1.22	0.71	0.59	17	44	39
	HD	1.22	0.69	0.56	15	48	37

¹ CON= basal diet with no additives; HD: basal diet with a mixture of 0.2% “Kräuterkraft Aufzucht & Verdauung”.

of the m^{th} lesion score at weaning ($m = 0, 1, 2$); WW_o was weaning weight as a covariate (linear regression); age_p was the age of the pig at the recording date as a covariate (linear regression); b_1 and b_2 were the regression coefficients of the trait on WW_o and age_p , respectively; S_q was the random effect of the q^{th} sire family ($q = 1$ –14); $S_q \times T_k$ was the effect of the interaction between sire and treatment, and $e_{ijklmnopqr}$ was the random residual effect.

Generalized linear mixed models with a logit link function from the “ordinal” package in R (Christensen, 2019) were applied to analyze the categorical trait LS from the post-weaning weight date. The respective model 2 was:

$$\begin{aligned} \text{logit}(P(y_{ijklmnopqr} \leq c)) &= \log \frac{P(y_{ijklmnopqr} \leq c)}{1 - P(y_{ijklmnopqr} \leq c)} \\ &= \mu_c + L_i + G_j + T_k + LN_l + SL_m + YS_n + b_1(WW_o) \\ &\quad + b_2(age_p) + S_q + (S_q \times T_k) \end{aligned} \quad (2)$$

where $y_{ijklmnopqr}$ was the observation for LS with three categories; $P(y_{ijklmnopqr} \leq c)$ was the cumulative probability of y less than or equal to a specific category $c = 1$ and 2, because $P(y_{ijklmnopqr} \leq 3) = 1$ and $1 - P(y_{ijklmnopqr} \leq 3) = 0$, with a maximum of $c = 3$; μ_c was the baseline value of the transformed cumulative probability for category c ; $S_q \times T_k$ was the interaction between sire and treatment. The remaining effects were the same as defined for model 1.

For both models 1 and 2, pairwise comparisons of least squares means were made using the Tukey test. Least square means with corresponding standard errors were considered statistically significant at a confidence level of 95% ($P \leq 0.05$). Spearman rank correlations were calculated between sire effect estimates for PWW, ADG and LS in the production environments CON and HD.

2.3.2. Models for genetic analyses

(Co)variance components were estimated using a Bayesian approach via Gibbs sampling. In this regard, we used the programs GIBBS3F90 and THRGIBBS3F90 from the BLUPF90 software package (Miszta et al., 2002). In genetic analyses, each trait recorded in the two different dietary treatment groups CON or HD were considered as two different traits in consecutive runs, e.g., PWW_{CON} and PWW_{HD} , ADG_{CON} and ADG_{HD} , and LS_{CON} and LS_{HD} . The bivariate animal model 3 was defined as follows:

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} X_1 & 0 \\ 0 & X_2 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} + \begin{bmatrix} Z_{a1} & 0 \\ 0 & Z_{a2} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} + \begin{bmatrix} Z_{c1} & 0 \\ 0 & Z_{c2} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} + \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} \quad (3)$$

where y_i were the observation vectors for the same trait (PWW, ADG or LS) recorded in the i^{th} dietary treatment ($i = 1$ and 2 for CON and HD, respectively); b_i were the vectors of fixed effects including sex, litter size, litter number, age of dam at farrowing and lesion score at weaning as categorical effects, and age at recording and weaning weight as covariables with the corresponding incidence matrix X_i ; a_i were the vectors of random additive genetic effects with incidence matrices Z_{ai} , and c_i were the vectors of random common litter environmental effects with incidence matrices Z_{ci} , and e_i were the vectors of random residual effects. The (co)variance structure for random effects in bivariate analyses was:

$$\text{var} \begin{bmatrix} a_1 \\ a_2 \\ c_1 \\ c_2 \\ e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} A\sigma_{a1}^2 & A\sigma_{a12} & 0 & 0 & 0 & 0 \\ A\sigma_{a12} & A\sigma_{a2}^2 & 0 & 0 & 0 & 0 \\ 0 & 0 & I\sigma_{c1}^2 & I\sigma_{c12} & 0 & 0 \\ 0 & 0 & I\sigma_{c12} & I\sigma_{c2}^2 & 0 & 0 \\ 0 & 0 & 0 & 0 & I\sigma_{e1}^2 & 0 \\ 0 & 0 & 0 & 0 & 0 & I\sigma_{e2}^2 \end{bmatrix}$$

where $\sigma_{a_i}^2$ and $\sigma_{c_i}^2$ were the additive genetic and the common litter

environmental variances, respectively, for the same traits in the i th dietary treatment ($i = 1$ and 2 for CON and HD, respectively); σ_{a12} and σ_{c12} were the additive genetic and common litter environmental covariances, respectively, between the same traits in CON and HD; σ_{e1}^2 and σ_{e2}^2 were the residual variances for the same traits in the respective treatment; A was the numerator relationship matrix among animals, and I were identity matrices for common litter environmental and residual effects. Heritability (h_i^2) and common litter environmental variance in relation to the phenotypic variance (c_i^2) for the trait in the i th treatment group were calculated as $h_i^2 = \sigma_{a_i}^2 / (\sigma_{a_i}^2 + \sigma_{c_i}^2 + \sigma_{e_i}^2)$, and $c_i^2 = \sigma_{c_i}^2 / (\sigma_{a_i}^2 + \sigma_{c_i}^2 + \sigma_{e_i}^2)$, respectively. The genetic correlation (r_a) and common litter environment correlation (r_c) between the same trait in CON and HD were calculated as $r_a = \sigma_{a12} / \sqrt{\sigma_{a1}^2 \times \sigma_{a2}^2}$ and $r_c = \sigma_{c12} / \sqrt{\sigma_{c1}^2 \times \sigma_{c2}^2}$, respectively. The same pig cannot be recorded in the same environment CON and HD. Consequently, there is no residual covariance between PWW_{CON} and PWW_{HD} , between ADG_{CON} and ADG_{HD} , and between LS_{CON} and LS_{HD} . Nevertheless, as implemented in the BLUPF90 software package (Misztal et al., 2002), we calculated an overall “total correlation” between same traits from different treatments considering the (co)variance components for additive genetic and common litter environmental effects as

$$r_t = \frac{(\sigma_{a12} + \sigma_{c12})}{\sqrt{(\sigma_{a1}^2 + \sigma_{c1}^2 + \sigma_{e1}^2) \times (\sigma_{a2}^2 + \sigma_{c2}^2 + \sigma_{e2}^2)}}, \text{ where } \sigma_{a12} \text{ and } \sigma_{c12} \text{ were the additive}$$

genetic and common litter environmental covariances between the same trait in CON and HD, respectively, and σ_{a1}^2 , σ_{c1}^2 and σ_{e1}^2 were the additive genetic, common litter environmental and residual variances for the i th dietary treatment.

In total, 200,000 Gibbs samples were generated, and 50,000 samples were discarded as a burn-in period. Afterward, a sampling interval of 100 cycles was considered to calculate posterior means and posterior standard deviations of (co)variance components. Statistics for post-Gibbs analyses were performed using the POSTGIBBSF90 program (Misztal et al., 2002). The length of the burn-in period and the number of iterations was determined based on the analyses of genetic covariances according to the method by Raftery and Lewis (1992), as implemented in the BOA software package (Smith, 2005).

3. Results

3.1. Fixed effects, random effects and sire x diet effects

Least squares means for ADG, PWW and LS with respective standard errors from model 1 for the effect levels are presented in Table 3. The season effect was significant ($P < 0.001$) for PWW, ADG and LS, indicating lower ADG and PWW for piglets born in spring and summer 2018 than for piglets born in winter from the same year. Age had a significant effect ($P < 0.001$) on ADG and PWW with least squares means of 5.51 g and 0.41 kg, respectively, (Table 3). The dietary treatment had no significant ($P > 0.05$) effect on PWW and ADG. Least squares means for PWW and ADG for animals from CON were larger than for animals from HD, but differences were not significant ($P > 0.05$).

The sex effect was not significant ($P > 0.05$) for LS at the post-weaning weight date. However, least squares means for LS were lower (= favorable) for female than for male piglets. Significant differences ($P < 0.05$) were observed between LS recorded in winter, spring and summer 2018. Least squares means for LS were highest in 2018-summer (1.55 ± 0.29), followed by spring (2018-spring: 0.54 ± 0.28 , 2019-spring: 0.80 ± 0.29). The trend for higher LS in the summer season was observed for each recording year. It was also observed that piglets with severe and mild lesions sores at the weaning date had increased least squares means for LS (1.54 and 0.83, respectively) at the end of the experiment at the post-weaning weight date. The dietary treatment significantly ($P < 0.05$) affected LS, indicating higher least squares means for LS for pigs allocated to the CON group (0.79) than for pigs

Table 3

Least square means and corresponding standard errors (in parenthesis) for levels of fixed effects on post weaning weight (PWW, in kg), average daily gain (ADG, in g) and lesion score (LS).

Effect	ADG	PWW	LS
Sex			
Male	402.00 (17.70)	21.20 (0.60)	0.72 (0.16)
Female	404.00 (17.80)	21.40 (0.60)	0.45 (0.17)
Year-Season			
2018-winter	431.00 ^{ab} (31.10)	22.40 ^{ab} (1.06)	-0.17 ^a (0.48)
2018-spring	379.00 ^{ab} (27.00)	20.70 ^{ab} (0.94)	0.54 ^a (0.28)
2018-summer	354.00 ^{ab} (48.50)	19.40 ^{ab} (1.66)	1.55 ^b (0.29)
2018-autumn	375.00 ^a (24.30)	20.80 ^{ab} (0.87)	0.60 ^{ab} (0.36)
2019-winter	452.00 ^b (23.20)	22.90 ^a (0.89)	0.23 ^{ab} (0.40)
2019-spring	426.00 ^{ab} (23.20)	21.50 ^b (0.83)	0.80 ^{ab} (0.29)
Age of animal	5.51 ^{***} (2.24)	0.41 ^{***} (0.08)	0.03 (0.06)
Litter number			
1	396.00 ^{ab} (18.70)	20.90 ^{ab} (0.64)	
2	385.00 ^a (18.70)	20.60 ^{ab} (0.64)	
3	399.00 ^{ab} (18.90)	21.10 ^{ab} (0.64)	
4	421.00 ^b (19.90)	21.90 ^b (0.68)	
5-7	414.00 ^{ab} (19.00)	21.80 ^b (0.64)	
Litter size			
≥12	434.00 ^a (21.30)	22.30 ^a (0.72)	0.56 (0.34)
13 - 15	400.00 ^{ab} (19.80)	21.50 ^{ab} (0.67)	0.43 (0.25)
16 - 17	389.00 ^b (18.20)	20.80 ^b (0.62)	0.65 (0.20)
18 - 19	397.00 ^{ab} (19.30)	21.00 ^{ab} (0.66)	1.04 (0.25)
≥20	395.00 ^{ab} (18.40)	20.90 ^b (0.62)	0.27 (0.17)
Weaning weight	18.42 ^{***} (2.44)	1.62 ^{***} (0.08)	0.30 ^{***} (0.07)
LS-W			
0	405.00 (18.90)	21.40 (0.64)	-0.59 ^a (0.21)
1	400.00 (17.90)	21.20 (0.61)	0.83 ^b (0.14)
2	404.00 (17.90)	21.30 (0.61)	1.54 ^c (0.27)
Group ²			
CON	408.00 (17.80)	21.50 (0.60)	0.79 ^a (0.17)
HD	398.00 (17.80)	21.10 (0.60)	0.39 ^b (0.16)

¹LS-W = lesion score recorded at weaning.

² CON= basal diet with no additives; HD: basal diet with a mixture of 0.2% “Kräuterkraft Aufzucht & Verdauung”. Within each effect, least square means with different letters differ significantly ($P \leq 0.01$). For regression coefficients, *** and ** represent significant at $P \leq 0.001$ and $P \leq 0.01$, respectively.

from the HD group. Pearson correlations between estimates for sire effects of same traits from both dietary groups CON and HD (i.e., the sire x diet effect) were 0.84 for PWW, 0.71 for ADG and 0.56 for LS.

3.2. Variance components and heritabilities in different dietary treatments

Variance components and corresponding variance ratios (heritabilities and the fraction of the common litter environment) for PWW, ADG and LS estimated in both groups CON and HD, are given in Table 4. Additive genetic variances for PWW were larger in the HD (2.94 ± 2.00) compared to CON group (2.27 ± 1.52). Accordingly, the estimated heritability for PWW in HD was slightly larger in CON than in HD (0.35 ± 0.16 vs 0.33 ± 0.14 , respectively). Similarly, a higher heritability was estimated for ADG in HD (0.34 ± 0.19) in comparison to CON (0.29 ± 0.14). In contrast, the heritability for LS was larger in CON than in HD (0.25 ± 0.14 vs 0.14 ± 0.12 , respectively). The common litter environmental variance as a proportion of the phenotypic variance ranged from 0.07 ± 0.05 for LS in HD to 0.16 ± 0.08 for PWW in HD.

3.3. Correlations between the same traits in different dietary treatments

The overall total correlations between the same traits in CON and HD were 0.36 ± 0.15 for PW, 0.33 ± 0.14 for ADG, and 0.10 ± 0.13 for LS (Table 5). Genetic correlations were 0.70 ± 0.18 between PWW_{CON} and PWW_{HD} , and 0.72 ± 0.19 between ADG_{CON} and ADG_{HD} , indicating genotype x diet interactions for both production traits. The small genetic correlation of 0.35 ± 0.20 between LS_{CON} and LS_{HD} indicates clear genotype x diet interactions. Correlations between common litter environmental effects were larger than the genetic correlations in same

Table 4

Posterior means (respective posterior SD in parentheses) of genetic parameters¹ for post weaning weight (PWW), average daily gain (ADG) and lesion score (LS) in both dietary groups.

Trait	Group ²	σ_p^2	σ_a^2	σ_c^2	σ_e^2	h^2	c^2
PWW	CON	8.11 (0.81)	2.77 (1.52)	0.98 (0.51)	4.35 (0.84)	0.33 (0.14)	0.12 (0.06)
	HD	8.19 (0.87)	2.94 (2.00)	1.33 (0.75)	3.92 (1.08)	0.35 (0.16)	0.16 (0.08)
ADG	CON	6589.8 (629.08)	1967.7 (1103.4)	672.87 (358.25)	3949.2 (638.20)	0.29 (0.14)	0.10 (0.05)
	HD	6433.5 (679.16)	2250.5 (1533.8)	922.46 (552.85)	3260.6 (840.97)	0.34 (0.19)	0.14 (0.08)
LS	CON	0.50 (0.06)	0.13 (0.08)	0.06 (0.03)	0.31 (0.05)	0.25 (0.14)	0.12 (0.06)
	HD	0.42 (0.04)	0.06 (0.05)	0.03 (0.02)	0.33 (0.04)	0.14 (0.12)	0.07 (0.05)

¹ σ_p^2 = phenotypic variance, σ_e^2 = residual variance, σ_a^2 = direct genetic variance, σ_c^2 = permanent environmental variance, c^2 = common litter environment, h^2 = direct heritability.

² CON = basal diet with no additives; HD = basal diet with a mixture of 0.2% “Kräuterkraft Aufzucht & Verdauung”.

traits, i.e., 0.74 ± 0.23 for PWW, 0.74 ± 0.24 for ADG and 0.60 ± 0.27 for LS.

Changes in estimated breeding values (EBV) of the 14 boars with offspring in CON and HD are presented in Fig. 1. For PWW, we only observed slight re-rankings of boars across the dietary groups. For instance, the sire 5 with an EBV of +0.63 kg was ranked on the third position in CON, but a decrease of 0.28 kg implied rank 5 in the HD environment. Similar results were observed for ADG, indicating that the same 5 sires were top-ranked in both treatment groups. In contrast, a considerable re-ranking of sires was observed according to their EBV for LS, displaying a shrinkage of EBV variations in the HD environment. The estimates of common litter environmental effects for 14 randomly selected dams in both groups CON and HD are presented in Fig. 1. Similar to the EBV of sires, only minor re-rankings were observed for the moderate heritability traits PWW and ADG. In contrast, for the functional trait LS, stronger dam re-rankings and a decline of effect variations in the HD group, were observed. However, the re-rankings and the variation shrinkage in the HD group were more obvious for the additive genetic component, i.e., for the sire EBVs.

4. Discussion

4.1. Herbal diet supplement effects on growth performance

Reports from the European Medicines Agency (2012) indicate that supplies with chicory roots prevent slow digestion, abdominal fullness and temporary loss of appetite in humans. Liu et al. (2011) and Velasco et al. (2010) also observed improvements in feed intake and body weight traits in broilers when adding chicory roots to the feed ration. However, in our trial, piglets from the CON group had 0.40 kg higher PWW and 10 g/day higher ADG compared to piglets from the HD group. In a pig feeding experiment including a group receiving an herbal diet, Presto et al. (2019) supported the results obtained from the present study. Similarly, Déru et al. (2020) identified better growth performances for pigs receiving a conventional diet compared to a “high fiber diet” group. In a pig feeding experiment conducted by Sevillano et al. (2018), the effect of the diet (i.e., cereal alternative ingredients versus a

corn-soybean diet) on growth parameters was non-significant. According to Lebre et al. (2018) and Liao et al. (2015), lysine supplements stimulated protein synthesis, and accordingly, a lysine and methionine deficiency was associated with limited growth. Nevertheless, in the present study, the percentages of lysine and methionine in the HD diet during the second phase of the dietary treatment were quite high, but ADG was lower than in the CON group. Ginane et al. (2015), Nyachoti et al. (2004) and de Haer et al. (1993) explained that lower ADG also depends on the physiological state of piglets and of growing pigs. Another possible explanation addresses the lower energy content of the HD treatment compared to the CON treatment (13.0 MJ/kg vs 14.1 MJ/kg). In this regard, several studies mentioned the lower energy content of diets enriched with fiber, with detrimental impact on growth parameters and feed efficiency (Cho et al., 2012; Li et al., 2012; Kallabis and Kaufmann, 2012). Martino et al. (2014) observed a negative effect of organic feeding characterized by higher roughage content on body weight traits in pigs, similarly to the results from the present study. However, in contrast, Zhao et al. (2016) associated supplementation of fermented herbs with improved growth performance and improved nutrient digestibility in weaning and post-weaning pigs. Wallenbeck et al. (2014) suggested only a small amount of crude fiber in the diet of growing pigs, to avoid negative effects on the digestive capacity.

4.2. Herbal diet supplement effect on lesion scores

Associations have been reported between weights of pigs and aggressiveness, with lighter pigs being more aggressive (Palander et al., 2013; Turner et al., 2008; Edwards, 2006; Van Erp-Van Der Kooij et al., 2003; Hessing et al., 1994) as identified in the present study. Although piglets in the HD group had more lesions at the weaning date indicating aggressiveness and increased activity in the pen, the same pigs had a smaller and favorable LS at the end of the experiment. Hence, their improved calmness could be due to the higher crude fiber (4.9% vs 2.4%) and crude protein (17.1% vs 16.6%) contents in the HD compared to the CON ration. The low LS in the HD group reflect the observations by Bernardino et al. (2016) and Kallabis and Kaufmann (2012), i.e., the favorable effects of diets enriched with fiber on social animal behavior, on feeding behavior and on overall calmness. In our feeding experiment, we made the observation in the HD group that pigs tend to spend more time to digest roughage, implying reduced time for foraging and improved wellbeing. Presto et al. (2019) and Carlson et al. (1999) observed limited abnormal behavior pattern such as nosing with other pigs and aggressiveness with group contemporaries when fed chicory, the active ingredient in the HD treatment. The chicory roots might stimulate calmness, and accordingly reduced fighting and aggressive behavior in the HD group, as proved by a fewer number of skin lesions in the HD than in the CON group. In addition, chicory herbs have medicinal properties to treat various ailments including diabetes and played a role in wound healing (Hakkinen et al., 2021; Saeed et al., 2017; Street et al., 2013), possibly explaining the lower and favorable LS in the HD group.

In the present study, the prevalence of skin lesions significantly ($P <$

Table 5

Overall total, genetic and common litter environmental covariances and correlations (respective posterior SD in parentheses) between the same traits (post weaning weight (PWW), average daily gain (ADG), lesion score (LS)) recorded in CON and HD groups¹.

Traits		Overall total	Genetic	Common litter
PWW	Covariance	3.11 (1.45)	5.12 (0.21)	0.87 (0.55)
	Correlation	0.36 (0.15)	0.70 (0.18)	0.74 (0.23)
ADG	Covariance	2190.8 (1062.5)	1591.3 (1219.9)	599.52 (391.68)
	Correlation	0.33 (0.14)	0.72 (0.19)	0.74 (0.24)
LS	Covariance	0.05 (0.06)	0.02 (0.06)	0.03 (0.02)
	Correlation	0.10 (0.13)	0.45 (0.20)	0.60 (0.27)

¹ CON = basal diet with no additives; HD = basal diet with a mixture of 0.2% “Kräuterkraft Aufzucht & Verdauung”.

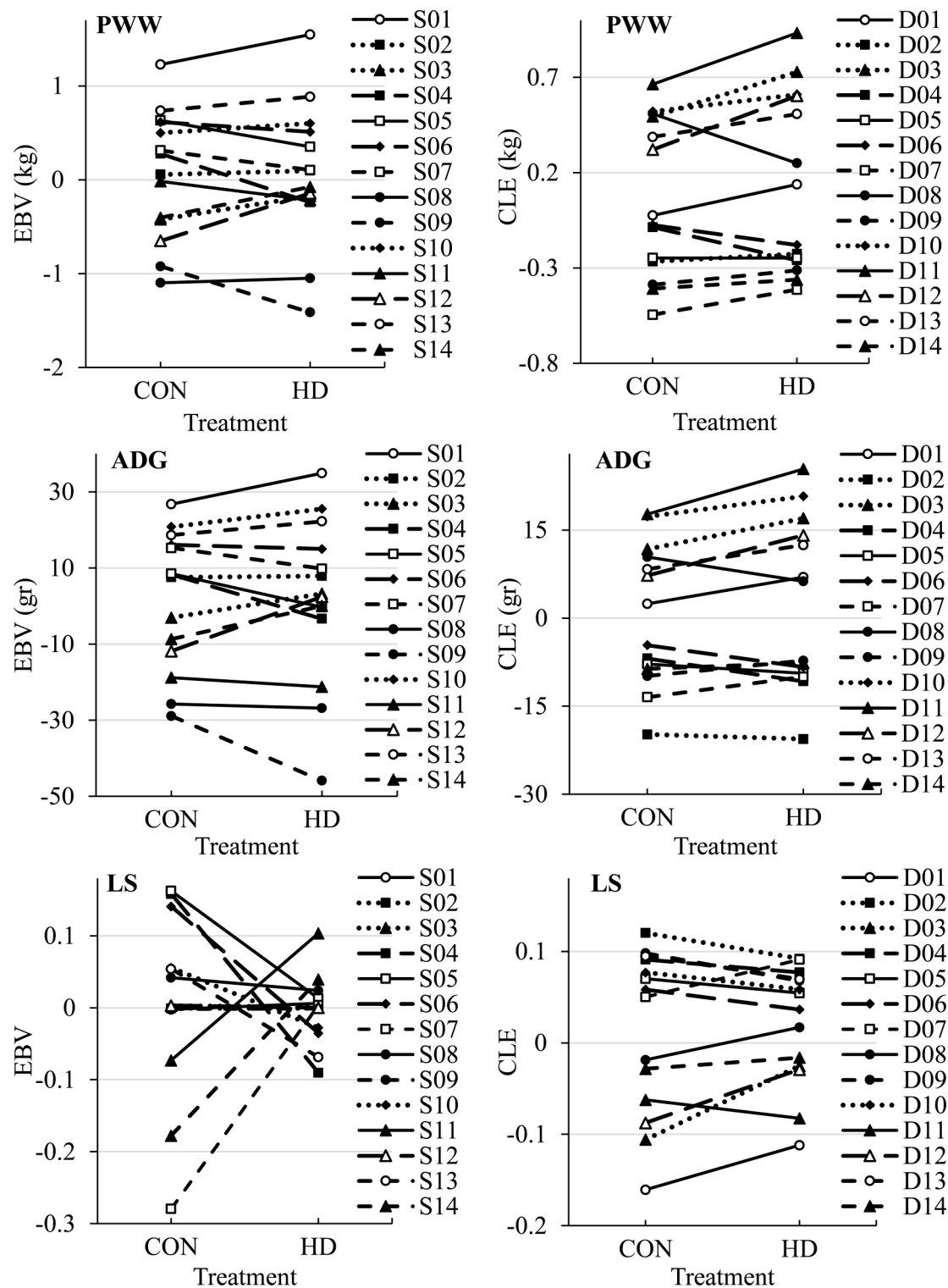


Fig. 1. Estimated breeding values (EBV) for the 14 sires (left) and common litter effects (CLE) for 14 randomly selected dams (right) for post weaning weight (PWW), average daily gain (ADG) and lesion score (LS) in CON (basal diet with no additives) and HD (basal diet with a mixture of 0.2% "Kräuterkraft Aufzucht & Verdauung").

0.001) increased with increasing body weight in the CON group. Camerlink et al. (2014) reported positive phenotypic correlations between skin lesions and body weight, and they assumed untypical behavior in extremely heavy pigs. In contrast, other researches (Turner et al., 2008; Edwards, 2006; Beattie et al., 2005; Schröder-Petersen and Simonsen, 2001) associated aggressive behavior and tail biting in pigs with small body sizes, but they did not study the relationship with skin

lesions.

4.3. Genetic parameters in different diet groups and diet x genotype interactions

The genetic statistical model selection based on the Akaike information criterion (AIC; Akaike (1973)), and the finally defined model 3

additionally considering the litter environmental effect displayed lower AIC than a model only including the random additive genetic effect. However, some of the genetic parameters were associated with quite large posterior SD, suggesting a larger sample size for this pig experiment. Accordingly, Gourdine et al. (2019) reported large posterior SD for genetic parameters in a backcross (Large White x Creole) pig population comprising 1298 piglets. Especially when fitting multiple-trait models for the proof of possible genotype x environment interactions (i.e., models without residual covariances), the accuracy of genetic parameter estimates strongly depended on the sample size (Falconer and Mackay, 1996).

The moderate heritabilities for PWW and ADG reflect estimates in other pig populations at a similar age (e.g. Zhang et al. 2016; Alves et al. 2018; Gourdine et al. 2019). Alves et al. (2018) estimated a direct heritabilities of 0.33 for PWW in Yorkshire and of 0.27 in Landrace pigs. Zhang et al. (2016) reported a heritability of 0.34 for ADG in a Landrace pig population in China, and Lopez et al. (2018) a heritability of 0.52 in a Landrace population located in Korea. For PWW and ADG, variance components and variance ratios were very similar in both dietary groups HD and CON. Genetic variances and heritabilities for PWW and ADG were only slightly larger in HD than in CON, reflecting a minor effect on feeding characteristics on genetic parameter estimates or on gene activities. Stronger alterations of genetic variances in dependency of dietary characteristics were observed for LS. The smaller genetic variance for LS in the HD group indicate intra-family similarity, and consequently, the similarity might contribute to narrowed genetic differentiation in the HD environment. The smaller variation of boar EBVs for LS in the HD group is a further indicator in this regard. For low heritability functional traits, Schierenbeck et al. (2011) suggested a “challenging test environment” to improve genetic differentiation. In contrast, for moderate heritability production traits, König et al. (2005) suggested to keep and to select animals in environments reflecting superior nutritional value, so that such an environment supports the fully expression of the true genetic potential. Nevertheless, from a practical breeding scheme perspective, the increased genetic variance, heritability and EBV range for LS simplify selection strategies on behavior in the challenging CON environment. In such context, Velie et al. (2009) and Rohrer et al. (2013) favored consideration of environmental stressors to optimize pig selection.

Interestingly, the common litter environmental effect explained about 12% of the phenotypic variation for the studied traits. In pig populations, families of full- and half-sibs share non-genetic litter-specific environmental conditions such as uterus capacity and nutrition during the suckling period. Thus, piglets sharing the same environment in early life develop similar body weight characteristics and social skills in adulthood (Kaufmann et al., 2008; Canario et al., 2017). Similarly, Canario et al. (2017) explained that social experience in early life strongly influences behavior patterns and social interactions of piglets in adulthood. Such physiological background may explain that the common litter environmental correlations between PWW_{CON} and PWW_{HD}, between ADG_{CON} and ADG_{HD} and between LS_{CON} and LS_{HD} were larger than the respective genetic correlation.

In several studies (e.g., König et al. 2005), a genetic correlation smaller than 0.80 between same traits recorded in different environments was considered as an indicator for genotype x environment interactions. Accordingly, genotype x diet interactions were proved in the present study for all traits PWW, ADG and LS. Especially for LS, the genetic correlation was extremely small (0.15). Accordingly, obvious and very significant sire x diet interactions were identified for LS. The correlations between estimates for boar effects in the HD and in the CON group were largest for PWW. For the moderate heritability growth rate, Hermes et al. (1999) disproved any genotype x feeding regime interactions, because the genetic correlation was close to one.

Nevertheless, in analogy to the results from the present study for PWW and ADG, Hermes (2004) identified re-rankings of sires in relation to feeding levels. Very obvious genotype x environment

interactions were reported by Godinho et al. (2018) when stratifying the environment according to feed content characteristics. Li and Hermes (2016) utilized mean performance levels of pigs as environmental descriptor in reaction norm models. In this regard, they identified significant genotype x environment interactions. Accordingly, Chiba et al. (2002) carried out a feeding experiment and identified variations of sire responses with environmental alterations. Hence, they suggested genotype dependent optimization of feeding strategies. Contrarily, Schiavon et al. (2019) and Sundrum et al. (2011) disproved breed or genotype x diet interactions. Fabian et al. (2002) proved feed by genotype interactions for the traits feed efficiency and weight gain in Duroc pigs. The major feed component contributing to significant interactions was the lysine content. As an explanation for genotype x feeding interactions, Elbert et al. (2020) addressed the specific “genetic make-up” of each sire. In summary, for production traits in pigs, most of the conducted research studies proved genotype x feeding interactions (Quander-Stoll et al., 2021; Hofer et al., 2018; Agroscope, 2017; Hamilton et al., 2003). Therefore, selection of appropriate boars implies consideration of the environmental particularities, especially the feeding strategies in different production systems.

5. Conclusion

The present study highlighted the importance of a specific diet, diet x sire and diet x genotype effects on performance traits and on skin lesions in pigs. The results showed that piglets allocated to the HD group had smaller LS, indicating the favorable effects of an herbal diet on skin lesions. In addition, the herbal treatment (chicory) did not have any negative effects on growth performances. The significant interactions between sire and treatment detected for ADG, PWW and LS indicate specific sire trait responses, depending on the dietary characteristics. Furthermore, via multiple-trait models, genetic correlations smaller than 0.80 between PWW_{CON} and PWW_{HD}, ADG_{CON} and ADG_{HD}, and LS_{CON} and LS_{HD} indicate genotype x diet interactions, especially for the functional and low heritability LS. Hence, re-rankings of sires according to their EBVs were observed in both environments CON and LD, displaying smaller genetic and EBV variations for LS in the HD group. The HD dietary supplement contributed to fewer skin lesions, probably due to the favorable effect on pig behavior, but a more pronounced genetic differentiation seems to be possible in the more challenging CON environment.

Author statement

Dear editor and reviewer, we are very happy about your positive comments. We made the few minor changes in the revised manuscript, which are highlighted in yellow.

Thanks and best regards.

Declaration of Competing Interest

We declare that we have no conflicts of interest.

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References

- Agroscope, 2017. Fütterungsempfehlungen Und Nährwerttabellen für Schweine [Feeding Recommendations and Nutrition Tables For pigs. 3rd edition], 3 ed. Lehrmittel Zentrale, Zollikofen, Switzerland.
- Aikins-Wilson, S., Bohloul, M., König, S., 2021. Maternal and direct genetic parameters for tail length, tail lesions, and growth traits in pigs. *J. Anim. Sci.* 99, 1–11. <https://doi.org/10.1093/jas/skaa398>.
- Akaike, H., 1973. Information theory and an extension of the maximum likelihood principle. In: Petrov, B.N., Csaki, F. (Eds.), *Proceedings of the 2nd International Symposium on Information Theory*. Akademiai Kiado.
- Alves, K., Schenkel, F.S., Brito, L.F., Robinson, A., 2018. Estimation of direct and maternal genetic parameters for individual birth weight, weaning weight, and probe weight in Yorkshire and Landrace pigs. *J. Anim. Sci.* 96, 2567–2578. <https://doi.org/10.1093/jas/sky172>.
- Augsburger, N.R., Ellis, M., Hamilton, D.N., Wolter, B.F., Beverly, J.L., Wilson, E.R., 2002. The effect of sire line on the feeding patterns of grow-finish pigs. *Appl. Anim. Behav. Sci.* 75, 103–114. [https://doi.org/10.1016/S0168-1591\(01\)00188-5](https://doi.org/10.1016/S0168-1591(01)00188-5).
- Bates, D., Mächler, M., Bolker, B.M., Walker, S.C., 2015. Fitting linear mixed-effects models using lme4. *J. Stat. Softw.* 67 <https://doi.org/10.18637/jss.v067.i01>.
- Beattie, V.E., Breuer, K., O'Connell, N.E., Sneddon, I.A., Mercer, J.T., Rance, K.A., Sutcliffe, M.E.M., Edwards, S.A., 2005. Factors identifying pigs predisposed to tail biting. *Anim. Sci.* 80, 307–312. <https://doi.org/10.1079/ASC40040307>.
- Bernardino, T., Tatemoto, P., Morrone, B., Rodrigues, P.H.M., Zanella, A.J., 2016. Piglets born from sows fed high fibre diets during pregnancy are less aggressive prior to weaning. *PLoS ONE* 11, 1–11. <https://doi.org/10.1371/journal.pone.0167363>.
- Bidanel, J.P., Ducos, A., 1996. Genetic correlations between test station and on-farm performance traits in Large White and French Landrace pig breeds. *Livest. Prod. Sci.* 45, 55–62. [https://doi.org/10.1016/0301-6226\(95\)00079-8](https://doi.org/10.1016/0301-6226(95)00079-8).
- Braund, J.P., Edwards, S.A., Riddoch, I., Buckner, L.J., 1998. Modification of foraging behaviour and pasture damage by dietary manipulation in outdoor sows. *Appl. Anim. Behav. Sci.* 56, 173–186. [https://doi.org/10.1016/S0168-1591\(97\)00100-7](https://doi.org/10.1016/S0168-1591(97)00100-7).
- Brouns, F., Edwards, S.A., English, P.R., 1994. Effect of dietary fibre and feeding system on activity and oral behaviour of group housed gilts. *Appl. Anim. Behav. Sci.* 39, 215–223. [https://doi.org/10.1016/0168-1591\(94\)90157-0](https://doi.org/10.1016/0168-1591(94)90157-0).
- Bünger, B., Schrader, L., Schrade, H., Zacharias, B., 2015. Agonistic behaviour, skin lesions and activity pattern of entire male, female and castrated male finishing pigs. *Appl. Anim. Behav. Sci.* 171, 64–68. <https://doi.org/10.1016/j.applanim.2015.08.024>.
- Camerlink, I., Arnott, G., Farish, M., Turner, S.P., 2016. Complex contests and the influence of aggressiveness in pigs. *Anim. Behav.* 121, 71–78. <https://doi.org/10.1016/j.anbehav.2016.08.021>.
- Camerlink, I., Bolhuis, J.E., Duijvestijn, N., Van Arendonk, J.A.M., Bijma, P., 2014. Growth performance and carcass traits in pigs selected for indirect genetic effects on growth rate in two environments. *J. Anim. Sci.* 92, 2612–2619. <https://doi.org/10.2527/jas.2013-7220>.
- Campbell, J.M., Crenshaw, J.D., Polo, J., 2013. The biological stress of early weaned piglets. *J. Anim. Sci. Biotechnol.* 4, 2–5. <https://doi.org/10.1186/2049-1891-4-19>.
- Canario, L., Lundeheim, N., Bijma, P., 2017. The early-life environment of a pig shapes the phenotypes of its social partners in adulthood. *Heredity (Edinb)* 118, 534–541. <https://doi.org/10.1038/hdy.2017.3>.
- Carlson, D., Lærke, H.N., Poulsen, H.D., Jørgensen, H., 1999. Roughages for growing pigs, With emphasis on chemical composition, Ingestion and faecal digestibility. *Acta Agric. Scand. A Anim. Sci.* 49, 129–136. <https://doi.org/10.1080/090647099424033>.
- Cheng, C., Zhang, X., Xia, M., Liu, Z., Wei, H., Deng, Z., Wang, C., Jiang, S., Peng, J., 2017. Effect of oregano essential oil and benzoic acid supplementation to a low-protein diet on meat quality, fatty acid composition, and lipid stability of longissimus thoracis muscle in pigs. *Lipids Health Dis.* 16, 1–9. <https://doi.org/10.1186/s12944-017-0535-1>.
- Chiba, L.L., Kuhlert, D.L., Frobish, L.T., Jungst, S.B., Huff-Lonergan, E.J., Lonergan, S.M., Cummins, K.A., 2002. Effect of dietary restrictions on growth performance and carcass quality of pigs selected for lean growth efficiency. *Livest. Prod. Sci.* 74, 93–102. [https://doi.org/10.1016/S0301-6226\(01\)00288-3](https://doi.org/10.1016/S0301-6226(01)00288-3).
- Cho, S.B., Han, I.K., Kim, Y.Y., Park, S.K., Hwang, O.H., Choi, C.W., Yang, S.H., Park, K.H., Choi, D.Y., Yoo, Y.H., 2012. Effect of lysine to digestible energy ratio on growth performance and carcass characteristics in finishing pigs. *Asian Australas. J. Anim. Sci.* 25, 1582–1587. <https://doi.org/10.5713/ajas.2012.12311>.
- Christensen, R.H.B., 2019. A tutorial on fitting cumulative link mixed models with clmm2 from the ordinal package. Available online at https://cran.r-project.org/web/packages/ordinal/vignettes/clmm2_tutorial.pdf.
- DEath, R.B., Arnott, G., Turner, S.P., Jensen, T., Lahrmann, H.P., Busch, M.E., Niemi, J.K., Lawrence, A.B., Sandoe, P., 2014. Injurious tail biting in pigs: how can it be controlled in existing systems without tail docking? *Animal* 8 (9), 1479–1497.
- De Cuyper, C., Tanghe, S., Janssens, S., Van den Broeke, A., Van Meensel, J., Aluwé, M., Ampe, B., Buys, N., Millet, S., 2019. The effect of Piétrain sire on the performance of the progeny of two commercial dam breeds: a pig intervention study. *Animal* 1–8. <https://doi.org/10.1017/S1757173119000429>.
- de Haer, L.C.M., Luiting, P., Aarts, H.L.M., 1993. Relations among individual (residual) feed intake, growth performance and feed intake pattern of growing pigs in group housing. *Livest. Prod. Sci.* 36, 233–253. [https://doi.org/10.1016/0301-6226\(93\)90056-N](https://doi.org/10.1016/0301-6226(93)90056-N).
- Déru, V., Bouquet, A., Hassenfratz, C., Blanchet, B., Carillier-Jacquin, C., Gilbert, H., 2020. Impact of a high-fibre diet on genetic parameters of production traits in growing pigs. *Animal* 14, 2236–2245. <https://doi.org/10.1017/S1757173120001275>.
- Edwards, S.A., 2006. Tail biting in pigs: understanding the intractable problem. *Vet. J.* 171, 198–199. <https://doi.org/10.1016/j.tvjl.2005.04.010>.
- EFSA, 2007. Scientific report on the risks associated with tail biting in pigs and possible means to reduce the need for tail docking considering the different housing and husbandry systems (question no. EFSA-Q-2006-013). Annex to the EFSA J. 611, 1–13.
- Elbert, K., Matthews, N., Wassmuth, R., Tetens, J., 2020. Effects of sire line, birth weight and sex on growth performance and carcass traits of crossbred pigs under standardized environmental conditions. *Arch. Anim. Breed.* 63, 367–376. <https://doi.org/10.5194/aab-63-367-2020>.
- Ellen, E.D., Rodenburg, T.B., Albers, G.A.A., Bolhuis, J.E., Camerlink, I., Duijvestijn, N., Knol, E.F., Muir, W.M., Peeters, K., Reimert, I., Sell-Kubiak, E., van Arendonk, J.A.M., Visscher, J., Bijma, P., 2014. The prospects of selection for social genetic effects to improve welfare and productivity in livestock. *Front. Genet.* 5, 1–14. <https://doi.org/10.3389/fgenet.2014.00377>.
- European Medicines Agency, 2012. Community herbal monograph on *Cichorium intybus* L., radix. EMA/HMPC/121816/2010. In: European Medicines Agency (EMA) and Committee on Herbal Medicinal Products (HMPC). London, UK.
- Fabian, J., Chiba, L.L., Kuhlert, D.L., Frobish, L.T., Nadarajah, K., Kerth, C.R., McElhenney, W.H., Lewis, A.J., 2002. Degree of amino acid restrictions during the grower phase and compensatory growth in pigs selected for lean growth efficiency. *J. Anim. Sci.* 80, 2610–2618. <https://doi.org/10.1093/ansci/80.10.2610>.
- Falconer, D.S., Mackay, T.F.C., 1996. *Introduction to Quantitative Genetics*, 4th ed. New York (NY): Longman Harlow.
- Fraser, D., Phillips, P.A., Thompson, B.K., Tennesen, T., 1991. Effect of straw on the behaviour of growing pigs. *Appl. Anim. Behav. Sci.* 30, 307–318. [https://doi.org/10.1016/0168-1591\(91\)90135-K](https://doi.org/10.1016/0168-1591(91)90135-K).
- Ginane, C., Bonnet, M., Baumont, R., Revell, D.K., 2015. Feeding behaviour in ruminants: a consequence of interactions between a reward system and the regulation of metabolic homeostasis. *Anim. Prod. Sci.* 55, 247–260. <https://doi.org/10.1071/AN14481>.
- Godinho, R.M., Bastiaansen, J.W.M., Sevillano, C.A., Silva, F.F., Guimaraes, S.E.F., Bergsma, R., 2018. Genotype by feed interaction for feed efficiency and growth performance traits in pigs. *J. Anim. Sci.* 96, 4125–4135. <https://doi.org/10.1093/jas/sky304>.
- Gourdine, J.L., Riquet, J., Rosé, R., Poulet, N., Giorgi, M., Billon, Y., Renaudeau, D., Gilbert, H., 2019. Genotype by environment interactions for performance and thermoregulation responses in growing pigs. *J. Anim. Sci.* 97, 3699–3713. <https://doi.org/10.1093/jas/skz245>.
- Häkkinen, S.T., Soković, M., Nohynek, L., Čirić, A., Ivanov, M., Stojković, D., Tsitko, I., Matos, M., Baixinho, J.P., Ivasiv, V., Fernández, N., Dos Santos, C.N., Oksman-Caldentey, K.M., 2021. Chicory extracts and sesquiterpene lactones show potent activity against bacterial and fungal pathogens. *Pharmaceuticals* 14. <https://doi.org/10.3390/ph14090941>.
- Hamilton, D.N., Ellis, M., Wolter, B.F., McKeith, F.K., Wilson, E.R., 2003. Carcass and meat quality characteristics of the progeny of two swine sire lines reared under differing environmental conditions. *Meat Sci.* 63, 257–263. [https://doi.org/10.1016/S0309-1740\(02\)00079-7](https://doi.org/10.1016/S0309-1740(02)00079-7).
- Hermesch, S., 2004. Genetic improvement of lean meat growth and feed efficiency in pigs. *Aust. J. Exp. Agric.* 44, 383–391. <https://doi.org/10.1071/ea04017>.
- Hermesch, S., Luxford, B., Graser, H., 1999. Genetic parameters for performance traits of pigs recorded under ad libitum and restricted feeding. *Proc. Assoc. Advmt. Anim. Breed. Genet.* 13, 142–145.
- Hessing, M.J.C., Scheepens, C.J.M., Schouten, W.G.P., Tielen, M.J.M., Wiekema, P.R., 1994. Social rank and disease susceptibility in pigs. *Vet. Immunol. Immunopathol.* 43, 373–387. [https://doi.org/10.1016/0165-2427\(94\)90158-9](https://doi.org/10.1016/0165-2427(94)90158-9).
- Hofer, A., Spring, P., Stoll, P., 2018. Wie Füttern Wir Unsere Prüftiere künftig? [How We Will Feed Our Test Animals in the future?]. *Suisseporcs information*.
- Holling, C., Beilage, E.G., Vidondo, B., Nathues, C., 2017. Provision of straw by a foraging tower - Effect on tail biting in weaners and fattening pigs. *Porc. Health Manag.* 3, 1–14. <https://doi.org/10.1186/s40813-017-0052-7>.
- Ivarsson, E., Frankow-Lindberg, B.E., Andersson, H.K., Lindberg, J.E., 2011. Growth performance, digestibility and faecal coliform bacteria in weaned piglets fed a cereal-based diet including either chicory (*Cichorium intybus* L) or ribwort (*Plantago lanceolata* L) forage. *Animal* 5, 558–564. <https://doi.org/10.1017/S17571731110002193>.
- Jensen, M.B., Kyriazakis, I., Lawrence, A.B., 1993. The activity and straw directed behaviour of pigs offered foods with different crude protein content. *Appl. Anim. Behav. Sci.* 37, 211–221. [https://doi.org/10.1016/0168-1591\(93\)90112-3](https://doi.org/10.1016/0168-1591(93)90112-3).
- Jensen, M.B., Pedersen, L.J., 2010. Effects of feeding level and access to rooting material on behaviour of growing pigs in situations with reduced feeding space and delayed feeding. *Appl. Anim. Behav. Sci.* 123, 1–6. <https://doi.org/10.1016/j.applanim.2009.12.015>.
- Kallabis, K.E., Kaufmann, O., 2012. Effect of a high-fibre diet on the feeding behaviour of fattening pigs. *Arch. Anim. Breed.* 55, 272–284. <https://doi.org/10.5194/aab-55-272-2012>.
- Kaufmann, D., Hofer, A., Bidanel, J.-P., Kunzi, N., 2008. Genetic parameters for individual birth and weaning weight and for litter size of Large White pigs. *J. Anim. Breed. Genet.* 117, 121–128. <https://doi.org/10.1046/j.1439-0388.2000.00238.x>.
- König, S., Dietl, G., Raeder, L., Swalve, H.H., 2005. Genetic relationships for dairy performance between large-scale and small-scale farm conditions. *J. Dairy Sci.* 88, 4087–4096.
- Lau, Y.Y.W., Pluske, J.R., Fleming, P.A., 2015. Does the environmental background (intensive v. outdoor systems) influence the behaviour of piglets at weaning? *Animal* 9, 1361–1372. <https://doi.org/10.1017/S1757173115000531>.

- Lebret, B., Batonon-Alavo, D.I., Perruchot, M.H., Mercier, Z., Gondret, F., 2018. Improving pork quality traits by a short-term dietary hydroxy methionine supplementation at levels above growth requirements in finisher pigs. *Meat Sci.* 145, 230–237. <https://doi.org/10.1016/j.meatsci.2018.06.040>.
- Lecce, J.G., Armstrong, W.D., Crawford, P.C., Ducharme, G.A., 1979. Nutrition and management of early weaned piglets: liquid vs dry feeding. *J. Anim. Sci.* 48, 1007–1014.
- Li, L., Hermes, S., 2016. Evaluation of sire by environment interactions for growth rate and backfat depth using reaction norm models in pigs. *J. Anim. Breed. Genet.* 133, 429–440. <https://doi.org/10.1111/jbg.12207>.
- Li, P., Zeng, Z., Wang, D., Xue, L., Zhang, R., Piao, X., 2012. Effects of the standardized ileal digestible lysine to metabolizable energy ratio on performance and carcass characteristics of growing-finishing pigs. *J. Anim. Sci. Biotechnol.* 3, 9. <https://doi.org/10.1186/2049-1891-3-9>.
- Liao, S.F., Wang, T., Regmi, N., 2015. Lysine nutrition in swine and the related monogastric animals: muscle protein biosynthesis and beyond. *Springerplus* 4, 1–12. <https://doi.org/10.1186/s40064-015-0927-5>.
- Liu, H.W., Tong, J.M., Zhou, D.W., 2011. Utilization of Chinese herbal feed additives in animal production. *Agric. Sci. China* 10, 1262–1272. [https://doi.org/10.1016/S1671-2927\(11\)60118-1](https://doi.org/10.1016/S1671-2927(11)60118-1).
- Liu, H.Y., Hou, R., Yang, G.Q., Zhao, F., Dong, W.G., 2018. *In vitro* effects of inulin and soya bean oligosaccharide on skatole production and the intestinal microbiota in broilers. *J. Anim. Physiol. Anim. Nutr. (Berl.)* 102, 706–716. <https://doi.org/10.1111/jpn.12830>.
- Lopez, B.I.M., Song, C., Seo, K., 2018. Genetic parameters and trends for production traits and their relationship with litter traits in Landrace and Yorkshire pigs. *Anim. Sci. J.* 89, 1381–1388. <https://doi.org/10.1111/asj.13090>.
- Martino, G., Mugnai, C., Compagnone, D., Grotta, L., Del Carlo, M., Sarti, F., 2014. Comparison of performance, meat lipids and oxidative status of pigs from commercial breed and organic crossbreed. *Animal* 4, 348–360. <https://doi.org/10.1039/ani4020348>.
- Misztal, I., Tsuruta, S., Strabel, T., Auvray, B., Druet, T., Lee, D.H., 2002. BLUF90 and related programs (BGF90). In: *Proceedings of the 7th world congress genetics applied to livestock production*. Montpellier.
- Nwafor, I.C., Shale, K., Achilonu, M.C., 2017. Chemical composition and nutritive benefits of chicory (*cichorium intybus*) as an ideal complementary and/or alternative livestock feed supplement. *Sci. World J.* <https://doi.org/10.1155/2017/7343928>, 2017.
- Nyachoti, C.M., Zijlstra, R.T., De Lange, C.F.M., Patience, J.F., 2004. Voluntary feed intake in growing-finishing pigs: a review of the main determining factors and potential approaches for accurate predictions. *Can. J. Anim. Sci.* 84, 549–566. <https://doi.org/10.4141/A04-001>.
- Palander, P.A., Heinonen, M., Simpura, I., Edwards, S.A., Valros, A.E., 2013. Jejunal morphology and blood metabolites in tail biting, victim and control pigs. *Animal* 7, 1523–1531. <https://doi.org/10.1017/S1751731113000669>.
- Pluske, J.R., Williams, I.H., 1996. Reducing stress in piglets as a means of increasing production after weaning: administration of amperozide or co-mingling of piglets during lactation? *Anim. Sci.* 62, 121–130. <https://doi.org/10.1017/S1357729800014405>.
- Presto Åkerfeldt, M., Nihlstrand, J., Neil, M., Lundeheim, N., Andersson, H.K., Wallenbeck, A., 2019. Chicory and red clover silage in diets to finishing pigs—Influence on performance, time budgets and social interactions. *Org. Agric.* 9, 127–138. <https://doi.org/10.1007/s13165-018-0216-z>.
- Quander-Stoll, N., Früh, B., Bautze, D., Zollitsch, W., Leiber, F., Scheeder, M.R.L., 2021. Sire-feed interactions for fattening performance and meat quality traits in growing-finishing pigs under a conventional and an organic feeding regimen. *Meat Sci.* 179, 108555. <https://doi.org/10.1016/j.meatsci.2021.108555>.
- Raftery, A.E., Lewis, S., 1992. Bernardo, J.M., Berger, J.O., Dawid, A.P., Smith, A.F.M. (Eds.), *How Many Iterations in the Gibbs sampler?* Pages 763–774 in *Bayesian Statistics 4*. Oxford University Press, Oxford, UK.
- R Core Team, 2020. R: A language and Environment for Statistical Computing. Version 4.0.2. R Foundation for Statistical Computing, Vienna, Austria. <http://www.r-project.org/index.html>.
- Rohrer, G.A., Brown-Brandl, T., Rempel, L.A., Schneider, J.F., Holl, J., 2013. Genetic analysis of behaviour traits in swine production. *Livest. Sci.* 157, 28–37. <https://doi.org/10.1016/j.livsci.2013.07.002>.
- Saeed, M., Abd El-Hack, M.E., Alagawany, M., Arain, M.A., Arif, M., Mirza, M.A., Naveed, M., Chao, S., Sarwar, M., Sayab, M., Dhama, K., 2017. Chicory (*Cichorium intybus*) herb: chemical composition, pharmacology, nutritional and healthful applications. *Int. J. Pharmacol.* 13, 351–360. <https://doi.org/10.3923/ijp.2017.351.360>.
- Schiavon, S., Bona, M.D., Carcò, G., Sturaro, E., Gallo, L., 2019. Responses of pigs of different genotypes to a variation in the dietary indispensable amino acid content in terms of their growth and Carcass and meat quality traits. *Animals (Basel)* 31 (8), 508. <https://doi.org/10.3390/ani9080508>, 9.
- Schierenbeck, S., Reinhardt, F., Reents, R., Simianer, H., König, S., 2011. Identification of informative cooperator herds for progeny testing based on yield deviations. *J. Dairy Sci.* 94, 6143–6152.
- Schrey, L., Kemper, N., Fels, M., 2019. Behaviour and skin injuries of piglets originating from a novel group farrowing system before and after weaning. *Agriculture* 9. <https://doi.org/10.3390/agriculture9050093>.
- Schröder-Petersen, D.L., Simonsen, H.B., 2001. Tail biting in pigs. *Vet. J.* 162, 196–210. <https://doi.org/10.1053/tvjl.2001.0605>.
- Sevillano, C.A., Nicolaiciuc, C.V., Molist, F., Pijlman, J., Bergsma, R., 2018. Effect of feeding cereals-alternative ingredients diets or corn-soybean meal diets on performance and carcass characteristics of growing-finishing gilts and boars. *J. Anim. Sci.* 96 (11), 4780–4788. <https://doi.org/10.1093/jas/sky339>.
- Smith, B.J., 2005. Bayesian output analysis program (BOA). Version 1.1. User's manual.
- Street, R.A., Sidana, J., Prinsloo, G., 2013. *Cichorium intybus*: traditional uses, phytochemistry, pharmacology, and toxicology. Evidence-based Complement. Altern. Med. <https://doi.org/10.1155/2013/579319>, 2013.
- Sundrum, A., Aragon, A., Schulze-Langenhorst, C., Büttfering, L., Henning, M., Stalljohann, G., 2011. Effects of feeding strategies, genotypes, sex, and birth weight on carcass and meat quality traits under organic pig production conditions. *NJAS Wageningen. J. Life Sci.* 58, 163–172. <https://doi.org/10.1016/j.njas.2011.09.006>.
- Turner, P.S., 2011. Breeding against harmful social behaviours in pigs and chickens: state of the art and the way forward. *Appl. Anim. Behav. Sci.* 134, 1–9. <https://doi.org/10.1016/j.applanim.2011.06.001>.
- Turner, S.P., Roehe, R., D'Eath, R.B., Ison, S.H., Farish, M., Jack, M.C., Lundeheim, N., Rydhmer, L., Lawrence, A.B., 2009. Genetic validation of postmixing skin injuries in pigs as an indicator of aggressiveness and the relationship with injuries under more stable social conditions. *J. Anim. Sci.* 87, 3076–3082. <https://doi.org/10.2527/jas.2008-1558>.
- Turner, S.P., Roehe, R., Mekaw, W., Farnworth, M.J., Knap, P.W., Lawrence, A.B., 2008. Bayesian analysis of genetic associations of skin lesions and behavioral traits to identify genetic components of individual aggressiveness in pigs. *Behav. Genet.* 38, 67–75. <https://doi.org/10.1007/s10519-007-9171-2>.
- Van Erp-Van Der Kooij, E., Kuijpers, A.H., Van Eerdenburg, F.J.C.M., Tielen, M.J.M., 2003. Coping characteristics and performance in fattening pigs. *Livest. Prod. Sci.* 84, 31–38. [https://doi.org/10.1016/S0301-6226\(03\)00072-1](https://doi.org/10.1016/S0301-6226(03)00072-1).
- Vázquez-Gómez, M., García-Contreras, C., Astiz, S., Torres-Rovira, L., Fernández-Moya, E., Olivares, Á., Daza, A., Óvilo, C., González-Bulnes, A., Isabel, B., 2020. Piglet birthweight and sex affect growth performance and fatty acid composition in fatty pigs. *Anim. Prod. Sci.* 60, 573–583. <https://doi.org/10.1071/AN18254>.
- Velie, B.D., Maltecca, C., Cassidy, J.P., 2009. Genetic relationships among pig behavior, growth, backfat, and loin muscle area. *J. Anim. Sci.* 87, 2767–2773. <https://doi.org/10.2527/jas.2008-1328>.
- Velasco, S., Ortiz, L.T., Alzueta, C., Rebolé, A., Treviño, J., Rodríguez, M.L., 2010. Effect of inulin supplementation and dietary fat source on performance, blood serum metabolites, liver lipids, abdominal fat deposition, and tissue fatty acid composition in broiler chickens. *Poult. Sci.* 89, 1651–1662. <https://doi.org/10.3382/ps.2010-00687>.
- Vermeulen, L., Van Beirendonck, S., Bulens, A., Van Thielen, J., Driessen, B., 2016. Sire line of pigs affects weaning weight, growth performance, and carcass characteristics of offspring. *J. Anim. Sci.* 94, 4360–4368. <https://doi.org/10.2527/jas.2016-0591>.
- Wallenbeck, A., Rundgren, M., Presto, M., 2014. Inclusion of grass/clover silage in diets to growing/finishing pigs – influence on performance and carcass quality. *Acta Agric. Scand. Sect. A* 64, 145–153. <https://doi.org/10.1080/09064702.2015.1006668>.
- Wurtz, K.E., Siegfors, J.M., Bates, R.O., Ernst, C.W., Steibel, J.P., 2017. Estimation of genetic parameters for lesion scores and growth traits in group-housed pigs. *J. Anim. Sci.* 95, 4310–4317. <https://doi.org/10.2527/jas2017.1757>.
- Yan, L., Meng, Q.W., Kim, I.H., 2012. Effects of fermented garlic powder supplementation on growth performance, nutrient digestibility, blood characteristics and meat quality in growing-finishing pigs. *Anim. Sci. J.* 83, 411–417. <https://doi.org/10.1111/j.1740-0929.2011.00973.x>.
- Zhang, Z., Zhang, H., Pan, R.Y., Wu, L., Li, Y.L., Chen, Z.M., Cai, G.Y., Li, J.Q., Wu, Z.F., 2016. Genetic parameters and trends for production and reproduction traits of a Landrace herd in China. *J. Integr. Agric.* 15, 1069–1075. [https://doi.org/10.1016/S2095-3119\(15\)61105-4](https://doi.org/10.1016/S2095-3119(15)61105-4).
- Zhao, P., Li, H., Lei, Y., Li, T., Kim, S., Kim, I., 2016. Effect of fermented medicinal plants on growth performance, nutrient digestibility, fecal noxious gas emissions, and diarrhea score in weanling pigs. *J. Sci. Food Agric.* 96, 1269–1274. <https://doi.org/10.1002/jsfa.7217>.